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Information Reliability and Welfare: A Theory of Coarse Credit Ratings

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1 Big picture

- **Question.** Why do credit rating agencies (CRAs) use a small number of rating categories to describe credit quality that actually lies on a continuum, even though coarseness reduces the precision of information and can lower welfare?
- **Setting.** The paper is a theory paper. A firm has a risky investment project that requires outside finance. Project quality q is continuous. A CRA observes or learns the project's quality and reports a rating that both the firm and investors use.
- **Core friction.** A higher rating is socially useful because it improves the firm's investment choice, but it is also privately valuable to the issuer because it lowers financing costs and transfers wealth from new investors to old shareholders.
- **Main conceptual contribution.** The paper models credit ratings as a *cheap-talk* game in which the CRA puts positive weight on both the social value of accurate ratings and the issuing firm's desire for a higher rating. Coarseness then arises endogenously as the only credible way to communicate information without allowing excessive inflation.
- **Main theoretical result.** Continuous truthful ratings are impossible in equilibrium once the CRA's objective is tilted toward the issuer. Any two ratings that occur in equilibrium must imply quality inferences that are separated by at least a strictly positive amount.
- **Equilibrium structure.** There are multiple equilibria indexed by the number of rating categories n , up to some maximum N . In each equilibrium, the continuous support of project quality is partitioned into n intervals, and the CRA reports one rating per interval.
- **Main comparative-static result.** Ratings become *finer* when the CRA cares more about social value (α is larger) or when imprecision is more costly for investment efficiency; they become *coarser* when the CRA cares more about issuer shareholder wealth (β is larger) or when the private gain from ratings inflation is stronger.
- **Main competition result.** Competition among CRAs is not a cure-all. With two CRAs, each individual CRA's rating can become *more* coarse because each rating is less pivotal for the posterior belief of investors. Even so, total information and welfare can still rise because investors combine two ratings.
- **Main regulatory result.** Policies that reduce issuer-catering incentives—for example, forcing issuers to obtain and disclose all ratings they purchase—reduce coarseness. By contrast, mechanically tying regulatory benefits to rating labels can worsen coarseness.
- **Empirical connection.** The paper points to Tang (2009) as supportive evidence: when Moody's refined its rating grid in 1982, firms' investment became more sensitive to rating changes, consistent with the model's prediction that finer ratings improve real decisions.
- **Economic takeaway.** Coarse ratings are not explained here by technological inability. They are an equilibrium response to conflict of interest in the rating process.

2 Data and measurement (key objects)

2.1 Why this is a theory paper but still has “key objects”

- There is no sample, no regression panel, and no instrument in the empirical sense.
- The paper’s “data” are the model primitives: project quality, investment payoffs, financing terms, rating rules, posterior beliefs, and welfare.
- The measurement task is therefore conceptual: define exactly what the rating changes, what welfare means, and what object is made coarse in equilibrium.

Interpretation. The note keeps the same study-note structure as an empirical paper, but in a theory paper “measurement” means carefully defining the model objects whose movement the propositions explain.

2.2 Agents, timing, and information

- There is one firm with an investment project.
- There are competitive, risk-neutral outside investors who finance the project.
- There is a credit rating agency that can determine project quality and issue a rating.
- Project quality q is drawn from a continuous distribution with support $K \subseteq (Q_l, Q_h)$.
- The firm has no internal funds and must raise the full investment amount externally.

A crucial modeling choice is that the rating affects *two* things at once:

- the firm’s investment decision,
- and the price or repayment terms at which outside finance is raised.

Interpretation. This dual role is what creates the tension between rating accuracy and rating inflation.

2.3 Project payoff and the firm’s investment problem

Let $\Pi(I, q)$ denote the risky payoff from investment I when project quality is q . The firm’s net present value is

$$V(I, q) = \mathbb{E}[\Pi(I, q) - I].$$

The paper assumes:

- **Assumption 1.** $V(I, q)$ is concave in I and maximized at $I^n(\mathbb{E}[q])$.
- **Assumption 2.** The marginal project payoff is increasing in project quality, so better projects deserve higher investment.

These assumptions imply:

- the optimal investment level is increasing in expected quality,
- a more precise estimate of quality leads to more efficient investment,

- and the variance of posterior beliefs about q is socially costly.

Interpretation. The paper needs investment efficiency to depend on information precision; otherwise coarse ratings would not reduce welfare.

2.4 Financing terms and the wealth-transfer channel

Outside investors break even in expectation, so repayment terms satisfy

$$\mathbb{E}[D(\Pi(I, q), I, \hat{q})] = I,$$

where $\hat{q}$ is perceived quality and $D(\cdot)$ is the expected payment promised to new investors.

The paper defines the expected wealth transfer from new investors to existing shareholders as

$$T(\hat{q}, q) = \mathbb{E}[D(\Pi(I, q), I, \hat{q})].$$

Under Assumption 3, this transfer is

- increasing and concave in perceived quality $\hat{q}$,
- decreasing in true quality q ,
- and such that a better perceived rating can redistribute surplus toward existing shareholders.

Interpretation. This is the issuer's reason to prefer a higher rating, even if the higher rating is not more accurate.

2.5 The two value objects: social value and firm value

Suppose investors infer expected quality $\hat{q}(r)$ from rating r .

The **social value of the rating** is the NPV of the induced investment decision:

$$SV(\hat{q}(r), q) = \mathbb{E}[V(I^n(\hat{q}(r)), q)].$$

The **value of the rating to the firm** is

$$FV(\hat{q}(r), q) = \mathbb{E}[V(I^n(\hat{q}(r)), q) + I^n(\hat{q}(r)) - D(\Pi(I^n(\hat{q}(r)), q), I^n(\hat{q}(r)), \hat{q}(r))].$$

- SV is maximized by an *accurate* posterior, $\hat{q}(r) = q$.
- FV is maximized by an *inflated* posterior, $\hat{q}(r) > q$, because higher perceived quality cheapens external finance.

Interpretation. These two objects are the heart of the paper. The CRA is pulled between accuracy and inflation.

2.6 The CRA's objective and the bias parameter

The CRA maximizes a weighted average of social value and firm value:

$$Z(\hat{q}(r), q) = \alpha SV(\hat{q}(r), q) + \beta FV(\hat{q}(r), q),$$

with $\alpha > 0$ and $\beta > 0$.

The paper shows that this objective is concave in inferred quality and maximized at an inflated posterior:

$$h(q) \equiv \arg \max_{\hat{q}(r)} Z(\hat{q}(r), q) \geq q + \eta,$$

where $\eta > 0$ captures the minimum wedge between the socially accurate posterior and the CRA's privately preferred posterior.

Interpretation. η is the model's summary measure of conflict of interest.

2.7 Ratings, beliefs, and equilibrium objects

The CRA's strategy is a rating rule $\rho(r \mid q)$, and investors update beliefs using Bayes' rule. The posterior mean inferred from a rating is

$$\hat{q}(r) = \mathbb{E}[q \mid r].$$

The paper's definition of a **coarse** rating system is that there exists $\varepsilon > 0$ such that any two ratings used in equilibrium imply posterior means at least ε apart.

Interpretation. The key action is not the label itself, but the posterior mean and the investment/financing choice it induces.

2.8 Objects added in the competition extension

When there are two CRAs, A and B ,

- each observes a private noisy signal about project quality,
- the signals are conditionally independent,
- informative in the monotone-likelihood-ratio sense,
- and partial substitutes.

Investors then infer expected quality from the *pair* of ratings (r^A, r^B) .

Interpretation. Competition changes the precision of each CRA's rating and the total information available to investors.

3 Models and empirical specifications (all equations + interpretation)

3.1 Benchmark without credit ratings

Without a CRA, the firm chooses investment to maximize

$$V(I, q) = \mathbb{E}[\Pi(I, q) - I].$$

Because project quality is uncertain, the firm bases investment on the expected value of q . Under Assumptions 1 and 2:

- better expected quality leads to higher investment,
- but residual uncertainty about quality lowers NPV,
- so imperfect information is socially costly even before adding the rating agency conflict.

Interpretation. The no-rating benchmark tells you why information precision matters at all. The CRA can help by revealing information, but its conflict of interest limits how much it can reveal.

3.2 How a rating affects investment and financing

When investors infer $\hat{q}(r)$ from rating r , the firm invests

$$I = I^n(\hat{q}(r)).$$

The rating then affects welfare through two channels:

- **investment-efficiency channel:** a more accurate posterior makes I closer to the first-best choice,
- **wealth-transfer channel:** a higher posterior lets the firm raise funds on more favorable terms.

This produces the two objects already defined:

$$SV(\hat{q}(r), q) = \mathbb{E}\left[V(I^n(\hat{q}(r)), q)\right],$$

$$FV(\hat{q}(r), q) = \mathbb{E}\left[V(I^n(\hat{q}(r)), q) + I^n(\hat{q}(r)) - D(\Pi(I^n(\hat{q}(r)), q), I^n(\hat{q}(r)), \hat{q}(r))\right].$$

Interpretation. The same rating that improves real investment can also transfer wealth from buyers of the security to current shareholders. That is why the CRA's problem is not just about information revelation.

3.3 CRA objective and cheap-talk equilibrium

The CRA chooses ratings to maximize

$$Z(\hat{q}(r), q) = \alpha SV(\hat{q}(r), q) + \beta FV(\hat{q}(r), q),$$

where α measures concern for rating accuracy/social value and β measures concern for issuer shareholder wealth.

An equilibrium requires:

- a rating rule $\rho(r \mid q)$,
- incentive compatibility of the CRA's rating choice,
- and rational investor beliefs

$$\widehat{q}(r) = \mathbb{E}[q \mid r].$$

Because the CRA's payoff depends on the posterior generated by the rating rather than on the rating label itself, the problem is a *cheap-talk* game in the sense of Crawford and Sobel (1982).

Interpretation. If investors trusted every small change in the rating label as literal truth, the CRA would always shade upward. So a fully continuous rating system cannot be credible once $\beta > 0$.

3.4 Proposition 1: ratings must be coarse in equilibrium

Proposition 1. If r and r' are two ratings used in equilibrium, then

$$|\widehat{q}(r) - \widehat{q}(r')| \geq \eta > 0.$$

Economic intuition.

- Suppose ratings were continuous and investors interpreted them literally.
- Then a CRA that prefers some inflation would always want to report a slightly higher rating than the truth.
- The only way to restore credibility is to make categories wide enough that jumping to the next category overshoots the CRA's preferred inflation.

Interpretation. Coarseness is a commitment device. The CRA still wants inflation, but wide enough bins make small manipulations impossible.

3.5 Proposition 2: equilibrium ratings partition the quality space

Proposition 2. For every $n \leq N$, there exists an equilibrium with n distinct credit ratings $r_1, \dots, r_n$ such that the continuous support of q is partitioned into intervals

$$(Q_l, a_1], (a_1, a_2], \dots, (a_{n-1}, Q_h],$$

and the CRA reports one rating per interval.

The cutoffs satisfy boundary conditions

$$a_0 = Q_l, \quad a_n = Q_h,$$

and each interior cutoff solves an indifference condition: at the boundary, the CRA is just indifferent between the lower and upper adjacent rating categories.

What this means.

- The model does not produce one unique number of categories.
- Instead, it produces multiple equilibria with different levels of granularity.

- There is, however, an upper bound N on how many categories can be sustained.

Interpretation. The important qualitative result is not multiplicity per se. It is that all equilibria are discrete partitions of a continuum, and the maximal attainable refinement is finite once the CRA's objective is biased.

3.6 The paper's equilibrium selection intuition

Following the Crawford–Sobel logic, the paper emphasizes the equilibrium with the *most* informative communication among the available equilibria.

- More categories mean more precise posterior beliefs.
- More precise posterior beliefs mean more efficient investment.
- So the “best” equilibrium among the feasible cheap-talk equilibria is the one with the most rating buckets.

Interpretation. This is the paper's way of turning multiplicity into a useful comparative-static object: ask how the maximum feasible number of categories changes with incentives and market structure.

3.7 Closed-form example with risky debt financing

To get explicit formulas, the paper assumes project payoff

$$\Pi = \begin{cases} \Pi_h \equiv (a+1)I - b(I-q)^2 & \text{with probability } p \in (0,1), \\ \Pi_l \equiv I + cq - d & \text{with probability } 1-p. \end{cases}$$

The firm chooses investment to maximize

$$V(I, q) = \mathbb{E}[\Pi - I] = \mathbb{E}[p\{aI - b(I-q)^2\} + (1-p)(cq - d)].$$

The optimal investment rule becomes

$$I^n(\hat{q}) = \hat{q} + \frac{a}{2b}.$$

If project quality is uncertain, the NPV equals

$$\mathbb{E}[V(I^n(\mathbb{E}[q]), q)] = m \mathbb{E}[q] - pb \text{Var}(q) + \frac{pa^2}{4b} - (1-p)d,$$

where $m = pa + (1-p)c$.

Interpretation. The welfare loss from uncertainty appears directly as the $pb \text{Var}(q)$ term. That is why precision matters more when pb is larger.

3.8 Social value, firm value, and CRA bias in the closed-form example

Under the closed-form assumptions, the paper shows that the social value of a rating simplifies to

$$SV = -(\hat{q} - q)^2,$$

so it is maximized by truthful revelation.

The value to the firm simplifies to

$$FV = -\left(\hat{q} - q - \frac{c(1-p)}{2pb}\right)^2,$$

so it is maximized by upward inflation of the posterior.

The CRA objective becomes

$$Z = -\alpha(\hat{q} - q)^2 - \beta\left(\hat{q} - q - \frac{c(1-p)}{2pb}\right)^2.$$

Hence the CRA's preferred inflation is

$$\delta = \frac{\beta(1-p)c}{2pb(\alpha + \beta)}.$$

Interpretation. δ is the cleanest summary of how much the CRA wants to inflate ratings in the tractable example.

3.9 Corollary 1: explicit rating categories and the maximum number of buckets

With uniform project quality on (Q_l, Q_h) and debt financing, the maximum number of sustainable rating categories in the most informative equilibrium is

$$N^* = \left\lfloor \frac{\sqrt{1 + 2(Q_h - Q_l)/\delta} + 1}{2} \right\rfloor.$$

If there are n ratings, the cutoffs satisfy

$$a_i = Q_l + \frac{Q_h - Q_l}{n}i - 2i(n-i)\delta,$$

and the posterior mean within a bucket is simply the midpoint of its interval.

Interpretation.

- If $\delta = 0$, ratings can be arbitrarily fine.
- As δ rises, the maximum number of categories falls quickly.
- The fall is nonlinear: small increases in bias can sharply reduce granularity.

3.10 Figure 1's quantitative message

The paper uses Figure 1 to show how fast precision can collapse as the bias term rises.

- If $\delta = 0$, the rating can be continuous and welfare loss from rating-induced imprecision is zero.
- If δ rises to only 0.1% of the standard deviation of project quality, the maximum number of categories drops to 42 and the welfare cost becomes about 0.12% of the welfare cost with no ratings.
- At 1%, 5%, and 10% of the standard deviation, the maximum number of categories drops to 13, 6, and 4, respectively.
- The corresponding welfare costs rise to about 1.2%, 5.7%, and 11.3% of the no-rating welfare cost.

Interpretation. The paper's claim is not merely qualitative. It is that even modest conflicts of interest can generate a surprisingly coarse rating grid.

3.11 Proposition 3: what makes ratings finer or coarser

Proposition 3. In the most informative equilibrium, the number of rating categories increases with

- the weight α on social value,
- the sensitivity pb of project NPV to uncertainty,

and decreases with

- the weight β on issuer shareholder wealth,
- and the term $(1-p)c$, which captures how strongly better perceived quality redistributes surplus toward existing shareholders.

Economic interpretation.

- More concern for investors or real efficiency $\Rightarrow$ finer ratings.
- More issuer-catering or bigger financing-side rents from inflation $\Rightarrow$ coarser ratings.

The paper also uses these comparative statics to make two cross-market predictions:

- **Structured products** should have fewer rating categories than corporate bonds because the real-investment value of precise ratings is weaker.
- **Consumer credit scores** should be less coarse than corporate credit ratings because the rater's revenue is less tied to pleasing any one consumer.

3.12 Competition among CRAs

The paper next studies two ex ante identical CRAs, each receiving a private noisy signal about project quality. Investors condition on the pair of ratings (r^A, r^B) .

Lemma 1.

- Each CRA's rating remains coarse in equilibrium.
- There exist equilibria in which each CRA partitions its own signal into discrete intervals and reports one rating per interval.

Interpretation. Adding a second CRA does not remove the cheap-talk problem for either CRA individually.

3.13 Proposition 4: competition can increase coarseness

Proposition 4. The maximum number of ratings reported by each CRA with two CRAs is less than or equal to the maximum number of ratings reported by a single monopoly CRA. Yet welfare in the most informative equilibrium is higher with two CRAs than with one.

Economic intuition.

- With two CRAs, each CRA's rating has a smaller marginal effect on the posterior because investors also observe the other CRA's rating.
- To move investor beliefs by the same amount, each CRA must therefore use *wider* bins.
- So each rating becomes coarser.
- But because investors observe two ratings, the total information set is richer; effectively there are many joint rating buckets from the investors' point of view.

Interpretation. Competition can worsen the precision of each label while still improving total information and welfare.

3.14 Appendix A: why competition can have an inverted-U effect on welfare

Appendix A endogenizes the CRA objective by letting future business depend on reputation and current business depend on pleasing issuers.

In the appendix, the ratio of weights in the CRA objective satisfies

$$\frac{\alpha}{\beta} = \frac{\lambda \phi(n) C(n) f^2 k_1}{f^1 k_2},$$

where

- f^1 and f^2 are current and future rating fees,
- $\phi(n)$ is the CRA's survival probability,
- $C(n)$ is the number of candidate CRAs a firm considers,
- and n is the number of competing CRAs.

This delivers a non-monotone competition result:

- when the number of CRAs is small, more competition can raise α/β by strengthening reputational pressure,
- when the number of CRAs is already large, more competition can lower α/β by reducing survival and future rents, thereby strengthening issuer-catering.

Interpretation. The paper’s full view is not “competition is bad.” It is “competition can be good at first and bad later.”

3.15 Welfare and regulatory implications

Ratings coarseness lowers welfare because it reduces the precision of investment-relevant information.

The paper’s main regulatory implications are:

- **Good idea:** reduce the CRA’s incentive to please issuers. Requiring issuers to obtain and disclose all ratings they purchase can reduce ratings shopping and lower β .
- **Bad idea:** attach regulatory benefits mechanically to rating labels. If higher labels automatically trigger regulatory benefits, issuers care even more about getting inflated labels, which increases the divergence between social value and CRA incentives.

Interpretation. The paper wants regulation to target the *incentive to distort the signal*, not just the ex post “accuracy” of a fixed rating grid.

4 Identification and instruments (what makes the estimates credible)

4.1 What makes the comparison credible in a theory paper

- There is no empirical IV or natural experiment here.
- Credibility comes from a transparent mechanism design exercise: start from a standard investment-and-financing problem, add a CRA whose objective mixes investor and issuer interests, and solve the cheap-talk equilibrium.
- The paper then changes one feature at a time—the size of the incentive conflict, the number of CRAs, and the regulatory environment—to show how each affects ratings coarseness.

Interpretation. The paper is persuasive if the incentive conflict is the right primitive; it does not rely on reduced-form correlation.

4.2 Why reverse causality is not the issue

- In an empirical ratings paper, one would worry that financing costs, project quality, and ratings all affect one another simultaneously.
- Here that simultaneity is built directly into the model and solved as an equilibrium problem.

- The right question is not reverse causality but whether the model’s assumptions are strong enough to generate coarse communication in a disciplined way.

4.3 The core assumption that generates coarseness

The single most important assumption is that the CRA’s preferred inferred quality exceeds the truth:

$$h(q) \geq q + \eta, \quad \eta > 0.$$

Combined with

- concavity of the CRA objective in the inferred quality,
- rational investor beliefs,
- and a continuous underlying quality space,

this is enough to make literal truthful communication impossible and discrete partitions necessary.

Interpretation. The paper does not assume coarseness. It assumes a biased objective and then derives coarseness.

4.4 Why ratings are coarse but not systematically “wrong” in equilibrium

A subtle but important point in the paper is that coarseness does *not* mean investors are systematically fooled.

- Investors know the CRA’s equilibrium strategy.
- They rationally interpret each rating as a *range* of qualities.
- So the posterior mean associated with a rating is unbiased given the category.

Interpretation. The welfare loss comes from imprecision, not from investors naively believing inflated labels.

4.5 Why competition can worsen individual precision yet improve welfare

- Each CRA’s label is less informative when another label is present because each label is less pivotal.
- That makes each CRA’s equilibrium categories wider.
- But the pair of ratings together can still contain more information than a single rating.

Interpretation. This is one of the most interesting credibility checks in the paper: the coarseness result is not just about monopoly. It survives in competition, but the welfare conclusion becomes subtler.

4.6 What is assumed and what is derived

Assumed:

- project quality is continuous,
- the CRA can observe or infer quality,
- investors are rational,
- the CRA values both social precision and issuer wealth,
- the closed-form example uses risky debt financing and a uniform quality distribution,
- the competition extension imposes signal independence and substitutability.

Derived:

- ratings are coarse in every equilibrium,
- there is a finite upper bound on the number of categories,
- greater issuer-catering makes ratings coarser,
- each CRA's rating can become coarser under competition,
- and welfare consequences depend on both precision and market structure.

4.7 What the paper can and cannot distinguish

- The paper can explain why a coarse rating grid survives even without technological or legal limits on precision.
- It can distinguish between the effect of changing CRA incentives and the effect of changing the number of CRAs.
- It cannot tell us the real-world magnitude of α , β , η , or δ for a given agency or asset class.
- It also does not estimate the welfare cost of actual ratings practices in the data.

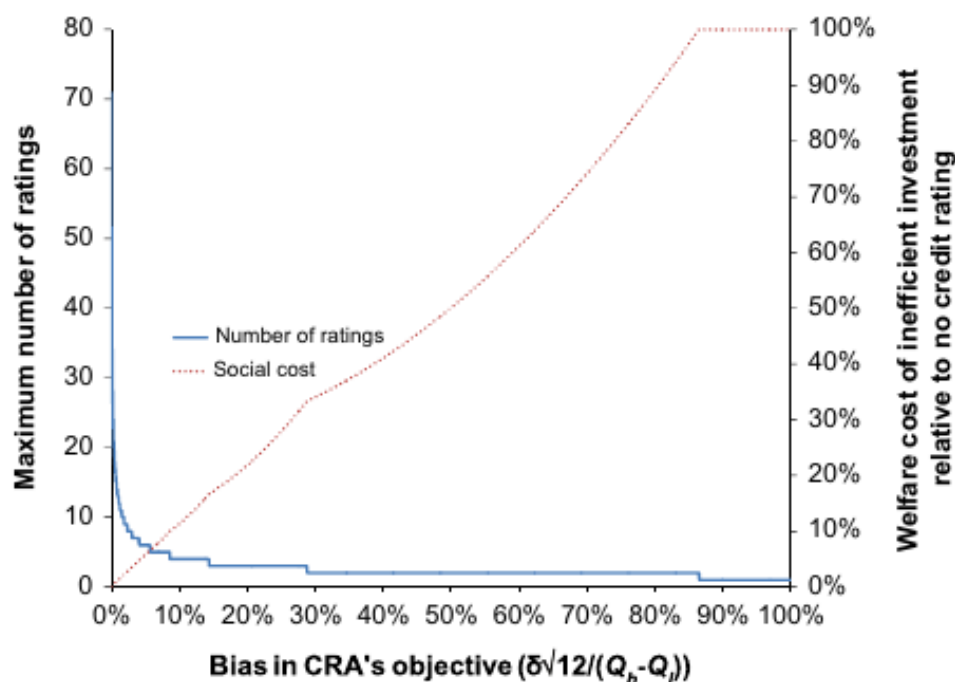
Interpretation. The paper is strongest as a mechanism paper, not as a calibrated quantitative assessment of modern ratings markets.

4.8 Relation to the literature

- Relative to Crawford and Sobel (1982), the contribution is to map a classic cheap-talk logic into the economics of credit ratings.
- Relative to ratings-shopping papers such as Bolton, Freixas, and Shapiro (2012), the paper focuses not on selective publication of ratings but on why the *rating scale itself* is coarse.
- Relative to the broader credit-rating literature, the paper adds a welfare result and a regulatory argument centered on information precision.

5 Tables and figures

5.1 Figure 1: bias in the CRA's objective and ratings coarseness



Read. The blue line shows the maximum number of rating categories. It falls sharply as the CRA's bias term δ rises as a fraction of the standard deviation of project quality. The red dotted line shows the welfare cost of inefficient investment relative to the no-rating benchmark, and it rises with δ .

Economic message. This figure is the cleanest visual summary of the paper. A stronger issuer-catering motive makes continuous communication impossible, compresses the rating grid, and sacrifices real efficiency.

5.2 What is missing from Section 5 relative to an empirical paper

- There are no regression tables in this paper.
- The paper is proposition-driven rather than table-driven.
- Figure 1 and the sequence of propositions are the core “evidence.”

6 Short summary

- **Question.** Why are credit ratings coarse when credit quality is continuous?
- **Mechanism.** The CRA values both rating accuracy and issuer shareholder wealth, so it wants to inflate ratings but not without bound.

- **Main result.** Truthful continuous communication is impossible; equilibrium ratings must be coarse.
- **Structure.** Project quality is partitioned into discrete intervals, one rating per interval, with a finite upper bound on the number of feasible categories.
- **Comparative statics.** More investor-oriented incentives make ratings finer; more issuer-oriented incentives make them coarser.
- **Competition.** A second CRA can make each rating more coarse, even though total information and welfare can still improve.
- **Policy implication.** Reduce ratings-shopping and issuer-catering incentives; do not mechanically attach more regulatory benefits to higher labels.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that ratings coarseness can emerge endogenously in a rational-expectations equilibrium. A CRA that cares both about social accuracy and about helping issuers raise cheap finance cannot credibly communicate a continuum of information. The equilibrium response is a finite set of rating categories.

7.2 Economic intuition

The intuition is simple and powerful. If every tiny upward change in a rating label were believed literally, the CRA would always want to exaggerate. Coarse categories solve this commitment problem because moving up one category becomes too large a distortion relative to the CRA's preferred amount of inflation. Ratings therefore remain informative, but only in chunks.

7.3 Takeaway

The paper's bottom line is that the coarseness of ratings is not just a descriptive curiosity. It is a consequence of incentive conflict, it affects real investment efficiency, and it can be altered by competition and regulation in non-obvious ways. In that sense, the rating scale itself is an equilibrium object that reflects the institutional design of the ratings industry.